Trøndelagkonkurranse 2026 - Level II

Second Round - November 1, 2026

1st Part

Choose and solve only one of the following problems:

Problem 1

Nils draws two different parallel lines A_1 and A_2 on the plane. The lines B and C each intersect both A_1 and A_2. Suppose that a line L satisfies the following conditions:

- L intersects A_1 at point P.
- L intersects A_2 at point S.
- L intersects B and C at the points Q and R.
- The order of the points P, Q, R and S in L is $P - Q - R - S$.

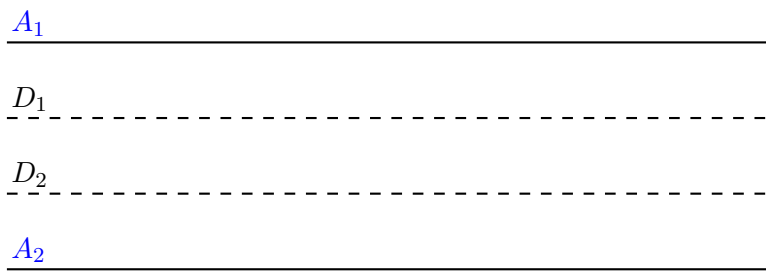
Show that Bianca can choose L such that the lengths of the segments PQ, QR and RS are all equal.

Problem 2

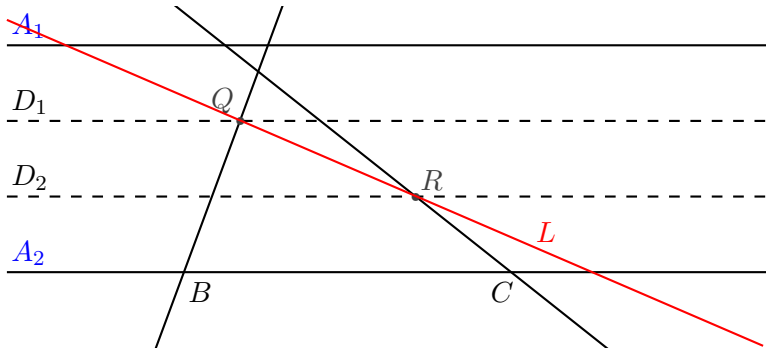
Let n be a positive integer. Each of the numbers $1, 2, 3, \ldots, n$ is written on a blackboard once. At each step, we can erase two numbers on the blackboard, say x and y, and replace them with the single number $|x - y|$. After several steps, there is only one number on the blackboard. For what values of n can this number be equal to 0?

Problem 1

Consider the lines D_1 and D_2 such that they lie between A_1 and A_2, are parallel to them, and divide the sector in between into three stripes of equal length.



Since B and C each intersect both A_1 and A_2, they also intersect D_1 and D_2. Thus, it suffices to consider $Q = D_1 \cap B$ and $R = D_2 \cap C$.



Thus, we can consider L passing through Q and R. Finally, any line that intersects A_1, D_1, D_2 and A_2 will produce three segments of equal length, and the result follows.

Problem 2

Let's first notice that the difference of even and odd values gives the output

$$\begin{aligned} |even - even| &= even \\ |odd - even| &= odd \\ |odd - odd| &= even \end{aligned}$$

If E is the amount of even numbers and O is the amount of odd numbers from 1 to n, then, after each step, applying the process of the problem reduces each quantity as

$$\begin{aligned} (E, O) &\to (E - 1, O) \\ (E, O) &\to (E - 1, O) \\ (E, O) &\to (E + 1, O - 2) \end{aligned}$$

This means that at all times, the amount of odd numbers has to be even. Since from 1 to $n = 4k$ we have an even amount of odd numbers, from 1 to $n = 4k + 1$ and from 1 to $n = 4k + 2$ we have an odd amount of odd numbers. This means that we can rule out $n = 4k + 1$ and $n = 4k + 2$.

Now we prove that $n = 4k+3$ and $n = 4k$ are the values we are looking for. Let's apply the process of the problem to $1, 2, 3, 4$. Taking 3 and 4, we obtain 1, so the sequence becomes

$$1, 2, 1$$

Taking 1 and 2, the sequence becomes

$$1, 1$$

which leads to 0 after one more step.

Let's now apply the process of the problem to $1, 2, 3$. Taking 3 and 2, we obtain

$$1, 1$$

which leads to 0 after one more step.

We show now that in general, any four consecutive values can be turned into zero. Consider $m, m+1, m+2, m+3$. Taking $m+3$ and $m+2$ we obtain

$$m, m+1, 1.$$

Taking m and $m+1$ we obtain

$$1, 1$$

which leads to 0 after one more step. Thus proving that we can add an arbitrary amount of blocks of $m, m+1, m+2, m+3$ to 1,2,3 and 1,2,3,4, so $n = 4k$ and $n = 4k+3$ are indeed allowed values for n.